

Department of Mechanical Engineering
Mid Semester Test II, November 2025
Fundamentals of Mechanical Engineering

Duration: 1:00 Hrs,
Attempt any two questions

Max Marks:20

- Q.1 (a) Define different types of links with examples. [CO 2, Marks 3]
- (b) Differentiate between machine and structure. [CO 2, Marks 3]
- (c) Three links are joined together to form a chain, show that it is not a kinematic chain. [CO 2, Marks 4]
- Q.2 (a) Determine mass density of 6 m^3 liquid volume having 44 kN weight. [CO 3, Marks 4]
- (b) Describe various types of fluid. [CO 3, Marks 4]
- (c) Define viscosity. [CO 3, Marks 2]
- Q.3 (a) List any five parts of I C engine and state their function. [CO 3, Marks 5]
- (b) Explain Otto cycle with the help P-V and T-S diagram and write expression of air standard efficiency. [CO 3, Marks 5]